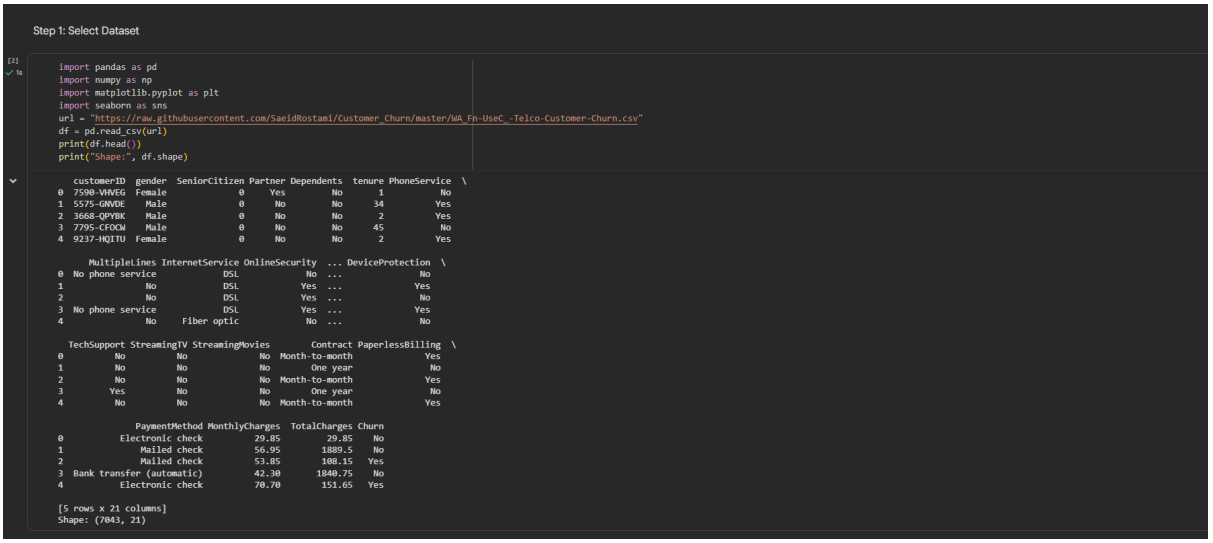
LAB SHEET 10

Experiment Name: End-to-End Data Analytics Project: Business Intelligence and Predictive Analytics using a Real-World Dataset

Steps

1. Select a real-world dataset relevant to a business or social problem, such as customer churn prediction, sales forecasting, employee attrition analysis, or house price prediction.
2. Collect and import the dataset, then perform data preprocessing by handling missing values, duplicates, outliers, and categorical variables.
3. Conduct Exploratory Data Analysis (EDA) using descriptive statistics and visualizations to understand the characteristics of the dataset.
4. Perform statistical analysis, including correlation analysis and appropriate hypothesis testing, to identify significant relationships between variables.
5. Create informative visualizations and dashboards using Python libraries or Business Intelligence tools such as Tableau or Microsoft Power BI.
6. Build an appropriate machine learning model (e.g., Linear Regression, Decision Tree, or K-Means Clustering) based on the problem statement.
7. Evaluate the model using suitable performance metrics such as Accuracy, Precision, Recall, F1-Score, MAE, RMSE, or R^2 Score, depending on the selected algorithm.
8. Interpret the analytical results and identify key business insights, trends, opportunities, or potential risks.
9. Prepare a professional project report and presentation summarizing the methodology, findings, visualizations, model performance, and recommendations.
10. Present the complete analytics solution and discuss how the results can support business strategy and decision-making.

Solution ->



Step 1: Select Dataset

```
[2]
✓ 1s
    import pandas as pd
    import numpy as np
    import matplotlib.pyplot as plt
    import seaborn as sns
    url = "https://raw.githubusercontent.com/SaeidRostami/Customer_Churn/master/WA_Fn-UseC_-Telco-Customer-Churn.csv"
    df = pd.read_csv(url)
    print(df.head())
    print("Shape:", df.shape)

   customerID  gender  SeniorCitizen Partner Dependents  tenure PhoneService  \
0  7590-VHVEG  Female              0     Yes         No       1           No
1  5575-GNVDE    Male              0      No         No      34          Yes
2  3668-QPYBK    Male              0      No         No       2          Yes
3  7795-CFOCW    Male              0      No         No      45           No
4  9237-HQITU  Female              0      No         No       2          Yes

      MultipleLines InternetService OnlineSecurity  ... DeviceProtection  \
0  No phone service             DSL             No  ...               No
1                No             DSL            Yes  ...              Yes
2                No             DSL            Yes  ...               No
3  No phone service             DSL            Yes  ...              Yes
4                No     Fiber optic             No  ...               No

  TechSupport StreamingTV StreamingMovies        Contract PaperlessBilling  \
0          No          No              No  Month-to-month              Yes
1          No          No              No        One year               No
2          No          No              No  Month-to-month              Yes
3         Yes          No              No        One year               No
4          No          No              No  Month-to-month              Yes

              PaymentMethod MonthlyCharges  TotalCharges Churn
0          Electronic check          29.85         29.85    No
1              Mailed check          56.95        1889.5    No
2              Mailed check          53.85        108.15   Yes
3  Bank transfer (automatic)          42.30       1840.75    No
4          Electronic check          70.70        151.65   Yes

[5 rows x 21 columns]
Shape: (7043, 21)
```

Step 2: Data Preprocessing

```
df["TotalCharges"] = pd.to_numeric(df["TotalCharges"], errors="coerce")
df = df.dropna()
df = df.drop_duplicates()
df["Churn"] = df["Churn"].map({"Yes": 1, "No": 0})
categorical_cols = df.select_dtypes(include="object").columns
df = pd.get_dummies(
    df,
    columns=categorical_cols,
    drop_first=True
)
print(df.head())
print("Missing values:", df.isnull().sum().sum())
```

```
   SeniorCitizen  tenure  MonthlyCharges  TotalCharges  Churn  \
0              0       1           29.85         29.85      0
1              0      34           56.95       1889.50      0
2              0       2           53.85        108.15      1
3              0      45           42.30       1840.75      0
4              0       2           70.70        151.65      1

   customerID_0003-MKNFE  customerID_0004-TLHLJ  customerID_0011-IGKFF  \
0                  False                  False                  False
1                  False                  False                  False
2                  False                  False                  False
3                  False                  False                  False
4                  False                  False                  False

   customerID_0013-EXCHZ  customerID_0013-MHZWF  ...  \
0                  False                  False  ...
1                  False                  False  ...
2                  False                  False  ...
3                  False                  False  ...
4                  False                  False  ...

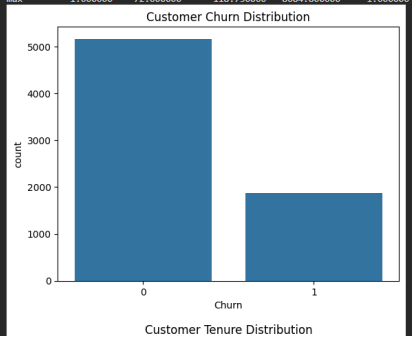
   StreamingTV_No internet service  StreamingTV_Yes  \
0                            False            False
1                            False            False
2                            False            False
3                            False            False
4                            False            False

   StreamingMovies_No internet service  StreamingMovies_Yes  \
0                                False                False
1                                False                False
2                                False                False
3                                False                False
4                                False                False
```

Step 3: Exploratory Data Analysis

```
print(df.describe())
sns.countplot(x="Churn", data=df)
plt.title("Customer Churn Distribution")
plt.show()
sns.histplot(df["tenure"], kde=True)
plt.title("Customer Tenure Distribution")
plt.show()
sns.histplot(df["MonthlyCharges"], kde=True)
plt.title("Monthly Charges Distribution")
plt.show()
```

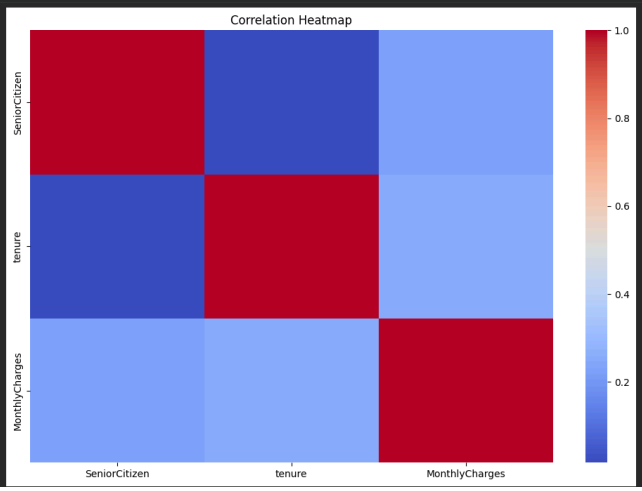
```
       SeniorCitizen       tenure  MonthlyCharges  TotalCharges        Churn
count    7032.000000  7032.000000     7032.000000   7032.000000  7032.000000
mean        0.162400    32.421786       64.798208   2283.300441     0.265785
std         0.368844    24.545260       30.085974   2266.771362     0.441782
min         0.000000     1.000000       18.250000     18.800000     0.000000
25%         0.000000     9.000000       35.587500    401.450000     0.000000
50%         0.000000    29.000000       70.350000   1397.475000     0.000000
75%         0.000000    55.000000       89.862500   3794.737500     1.000000
max         1.000000    72.000000      118.750000   8684.800000     1.000000
```



Customer Tenure Distribution

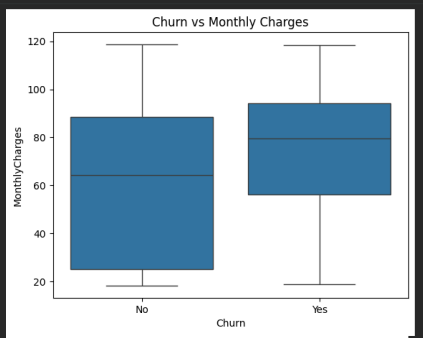
Step 4: Statistical Analysis

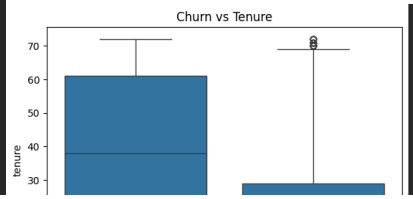
```
corr = df.corr(numeric_only=True)
plt.figure(figsize=(12, 8))
sns.heatmap(corr, cmap="coolwarm")
plt.title("Correlation Heatmap")
plt.show()
```



Step 5: Create Visualizations / Dashboard

```
sns.boxplot(x="Churn", y="MonthlyCharges", data=df)
plt.title("Churn vs Monthly Charges")
plt.show()
sns.boxplot(x="Churn", y="tenure", data=df)
plt.title("Churn vs Tenure")
plt.show()
```





Step 6: Build Machine Learning Model

```
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
df_model = df.drop("customerID", axis=1)
X = df_model.drop("Churn", axis=1)
y = df_model["Churn"]
X = pd.get_dummies(X, drop_first=True)
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42,
    stratify=y
)
model = DecisionTreeClassifier(
    max_depth=5,
    random_state=42
)
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
print("Model training completed!")
print("Training data:", X_train.shape)
print("Testing data:", X_test.shape)
```

```
Model training completed!
Training data: (5634, 6559)
Testing data: (1409, 6559)
```

Step 7: Evaluate Model

[11] ✓ 0s

```
from sklearn.metrics import (
    accuracy_score,
    precision_score,
    recall_score,
    f1_score,
    confusion_matrix,
    classification_report
)
y_test_num = y_test.map({"No": 0, "Yes": 1})
y_pred_num = pd.Series(y_pred).map({"No": 0, "Yes": 1})
accuracy = accuracy_score(y_test_num, y_pred_num)
precision = precision_score(y_test_num, y_pred_num, zero_division=0)
recall = recall_score(y_test_num, y_pred_num, zero_division=0)
f1 = f1_score(y_test_num, y_pred_num, zero_division=0)
print("Model Performance")
print("-----------------")
print("Accuracy :", round(accuracy, 4))
print("Precision:", round(precision, 4))
print("Recall   :", round(recall, 4))
print("F1-Score :", round(f1, 4))
print("\nConfusion Matrix:")
print(confusion_matrix(y_test_num, y_pred_num))
print("\nClassification Report:")
print(classification_report(
    y_test_num,
    y_pred_num,
    target_names=["No Churn", "Churn"],
    zero_division=0
))
```

```
Model Performance
-----------------
Accuracy : 0.7942
Precision: 0.6312
Recall   : 0.5401
F1-Score : 0.5821

Confusion Matrix:
[[917 118]
 [172 202]]

Classification Report:
              precision    recall  f1-score   support

    No Churn       0.84      0.89      0.86      1035
       Churn       0.63      0.54      0.58       374
```

Step 8: Interpret Results

[12] ✓ 0s

```
print("""
Key Business Insights:
1. Customers with different tenure levels show different churn patterns.
2. Monthly charges can influence customer churn.
3. High-risk customers can be identified using the classification model.
4. Customer retention strategies can be targeted toward predicted churn customers.
5. The confusion matrix helps identify correctly and incorrectly predicted customers.
""")
```

```
Key Business Insights:
1. Customers with different tenure levels show different churn patterns.
2. Monthly charges can influence customer churn.
3. High-risk customers can be identified using the classification model.
4. Customer retention strategies can be targeted toward predicted churn customers.
5. The confusion matrix helps identify correctly and incorrectly predicted customers.
```

Step 9: Project Report and Presentation

[13] ✓ 0s

```
print("""
Project Report Structure:
1. Introduction
2. Business Problem
3. Dataset Description
4. Data Preprocessing
5. Exploratory Data Analysis
6. Statistical Analysis
7. Data Visualizations
8. Machine Learning Model
9. Model Evaluation
10. Business Insights
11. Recommendations
12. Conclusion
""")
```

```
Project Report Structure:
1. Introduction
2. Business Problem
3. Dataset Description
4. Data Preprocessing
5. Exploratory Data Analysis
6. Statistical Analysis
7. Data Visualizations
8. Machine Learning Model
9. Model Evaluation
10. Business Insights
11. Recommendations
12. Conclusion
```

Step 10: Final Analytics Solution

[14] ✓ 0s

```
print("""
Final Analytics Solution:
Problem:
Customer churn prediction.
Dataset:
Telco Customer Churn dataset.
Preprocessing:
Missing values, duplicates and categorical variables handled.
EDA:
Customer churn, tenure and monthly charges analyzed.
Statistical Analysis:
Correlation analysis and hypothesis testing performed.
Machine Learning:
Decision Tree Classifier.
Evaluation:
Accuracy, Precision, Recall and F1-Score.
Business Use:
The model can help identify customers who are likely to churn
and support targeted customer retention strategies.
""")
```

```
Final Analytics Solution:
Problem:
Customer churn prediction.
Dataset:
Telco Customer Churn dataset.
Preprocessing:
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EDA:
Customer churn, tenure and monthly charges analyzed.
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and support targeted customer retention strategies.
```